BUILDING A USB RC METER

# Episode 1 Worksheet

## Why Build a Meter? How Instruments Measure R and C

*Work through these after watching the episode. Show your working; answers are on the last page. Values come from the 2014 thesis data unless stated.*

### Learning objectives checked

- Explain how a meter turns Ohm's law into a resistance reading
- Describe how measuring TIME can reveal capacitance
- Tell accuracy, precision, resolution and tolerance apart
- Read the block diagram of the USB RC meter built in this series

### Questions

1. A constant current of 1.5 mA flows through an unknown resistor and 2.25 V is measured across it. What is R?

2. In the divider method, $R_{REF}$ = 10 kΩ, $V_S$ = 5.0 V and $V_X$ = 2.0 V. Find $R_X$.

3. A capacitor is charged through 30 kΩ and reaches ⅔ of the supply after 3.3 s. Estimate C.

4. What range of values is allowed for a 4.7 kΩ ±5 % resistor?

5. Is 260 Ω an E24 value? Which E24 values are nearest?

6. A meter reads 102.0 Ω, 102.1 Ω, 101.9 Ω for a precision 100.00 Ω resistor. Is it precise, accurate, both or neither?

7. The CD800a is specified as ±(1.2 % + 5 digits) on its 4 kΩ range (1 Ω resolution). What is the uncertainty window for a 2.158 kΩ reading?

8. USB 2.0 allows VBUS between 4.75 V and 5.25 V at full load. What percentage spread is that around 5 V?

## Answers

1. R = V / I = 2.25 V / 1.5 mA = 1.5 kΩ.
2. $R_X = R_{REF} \cdot V_X / (V_S - V_X)$ = 10 kΩ × 2.0 / 3.0 = 6.67 kΩ.
3. C = t / (R · ln 3) = 3.3 / (30 000 × 1.0986) ≈ 100 µF.
4. 4.465 kΩ to 4.935 kΩ.
5. No. The nearest E24 values are 240 Ω and 270 Ω.
6. Precise (readings agree within ±0.1 Ω) but not accurate (about 2 % high): a calibration or offset error.
7. ±(0.012 × 2158 + 5) = ±30.9 Ω → about 2.127 to 2.189 kΩ.
8. ±0.25 V / 5 V = ±5 % (10 % total spread); down to 4.40 V it is −12 %.

### Going further

**USB in a NutShell, ch. 2 (VBUS limits):** https://www.beyondlogic.org/usbnutshell/usb2.shtml

**Sanwa CD800a specifications:** https://overseas.sanwa-meter.co.jp/products/digital_multimeters/cd800a.html

**AVR Transistortester (mikrocontroller.net):** https://www.mikrocontroller.net/articles/AVR_Transistortester

**Microsoft: end of support for Windows 10, 8.1 and 7:** https://www.microsoft.com/en-us/windows/end-of-support

**Plugable: 240 W USB PD EPR explained:** https://plugable.com/blogs/news/what-is-240w-usb-extended-power-range-epr

**Keysight: four-wire resistance measurement:** https://www.keysight.com/us/en/use-cases/measure-resistance-using-four-wire-measurement.html